P13: Building a Bee Monitoring Pipeline for Environmental Observation

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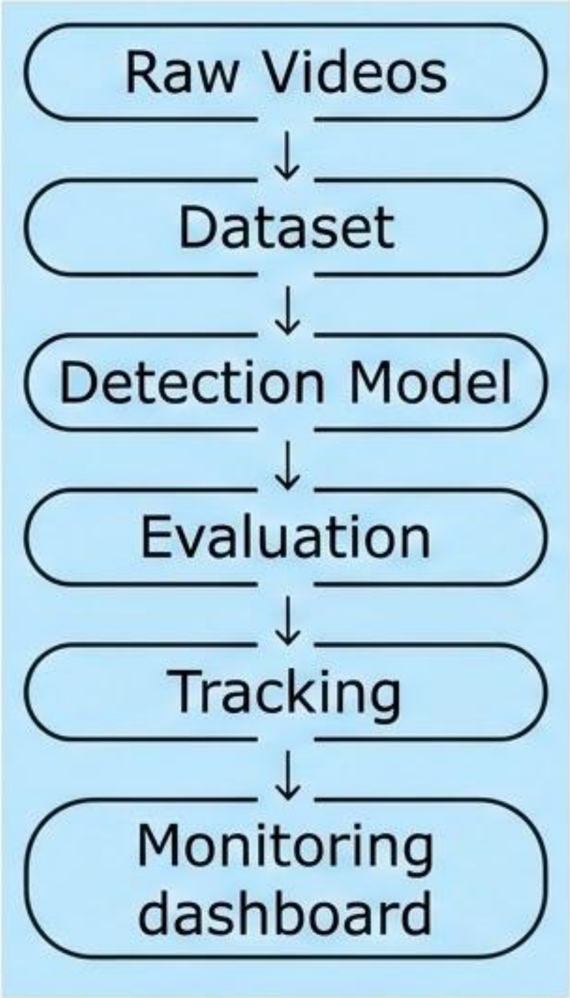
Project Overview



Objective :

- Develop an automated bee monitoring pipeline.
- Detect and classify pollinators from field videos.
- Compare state-of-the-art detection models.
- Build the foundation for future tracking and monitoring.

Pipeline Progress

📹BuzzSet dataset prepared and selected based on bee activity.✅

🧠Multiple YOLO variants trained on bee detection dataset. ✅

📊Model Evaluation completed using Precision, Recall and mAP metrics.✅

🔍False positive analysis completed. ✅

🎯RF-DETR baseline training in progress. 🔄

📍 Tracking pipeline integration in progress.🔄

📈Dashboard & monitoring statistics. ⏳

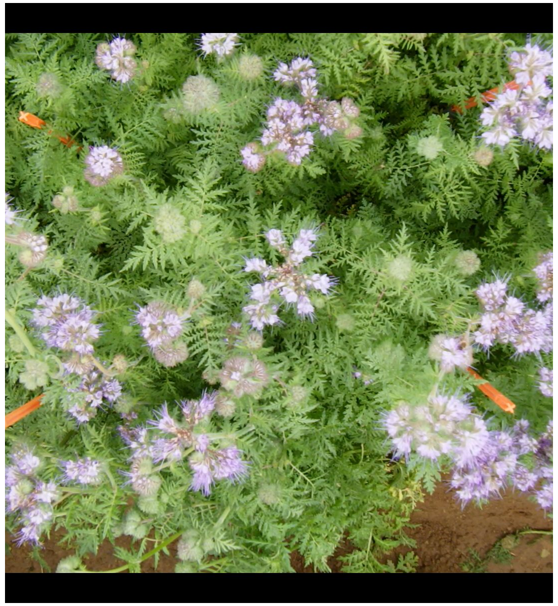
Dataset progress

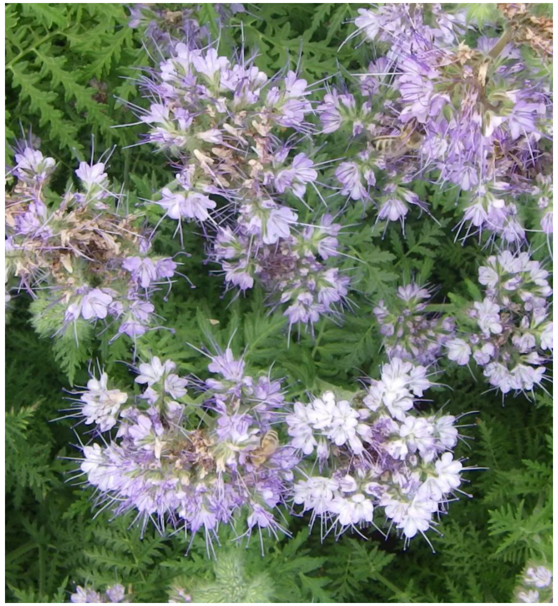
BuzzSetV2_split:

- valid:936
- train:5400
- test: 2155

2024 Videos:

- ~70 GB raw video
- 20240724 - 24 Videos
- 20240801 - 6 videos



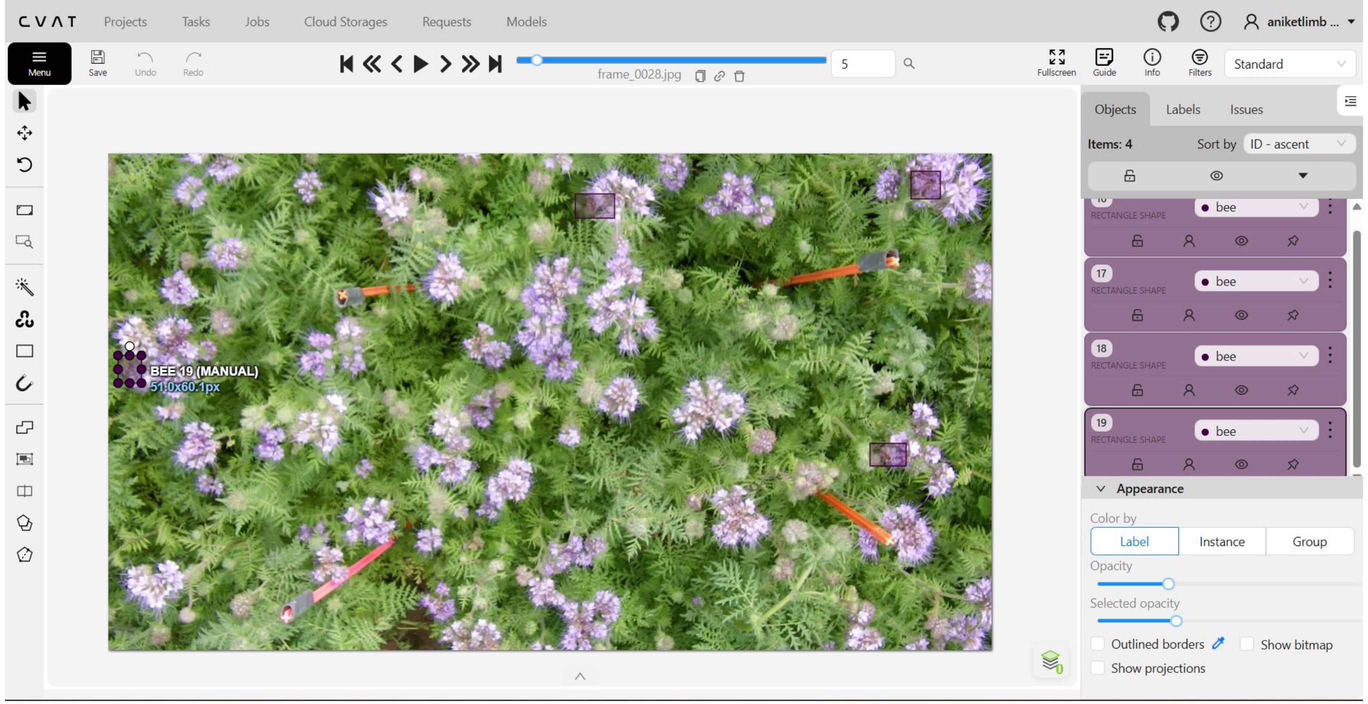
Dataset progress

ANNOTATION

- ~2,000 candidate frames extracted
- ~700 images annotated in CVAT
- ~1000 pollinators are labeled

DATASET MANAGEMENT

- Ultralytics YOLO dataset generated
- Train/Validation split completed
- Dataset expansion in progress



Detection model overview

YOLO Family (CNN-based)

Scale: **Nano → Small → Medium → Large → XL**

YOLO8

Mature & efficient

YOLO11

Speed + accuracy

YOLO26 Max capacity

RF-DETR (Transformer)

- End-to-end, no NMS
- Strong localization & small-object detection
- Higher accuracy, slower inference

Selected Baselines

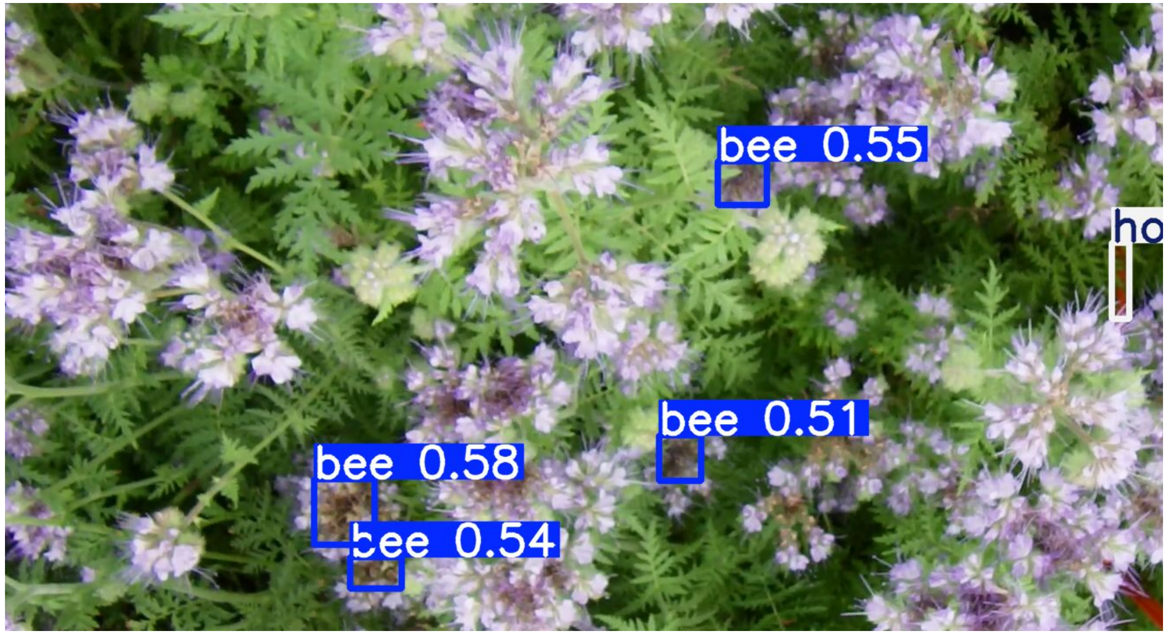
⚡ **YOLO11x**	🎯 **RF-DETR Large**
Real-time CNN baseline	Transformer accuracy baseline

Model Training

Metric	YOLO26X	YOLO11L	YOLOV8L	RF-DETR
mAP@50-95	**0.322**	**0.275**	**0.223**	**0.325**
mAP@50	**0.484**	**0.421**	**0.411**	**0.535**
Precision	**0.512**	**0.638**	**0.701**	**0.601**
Recall	**0.452**	**0.413**	**0.400**	**0.540**

YOLO 11

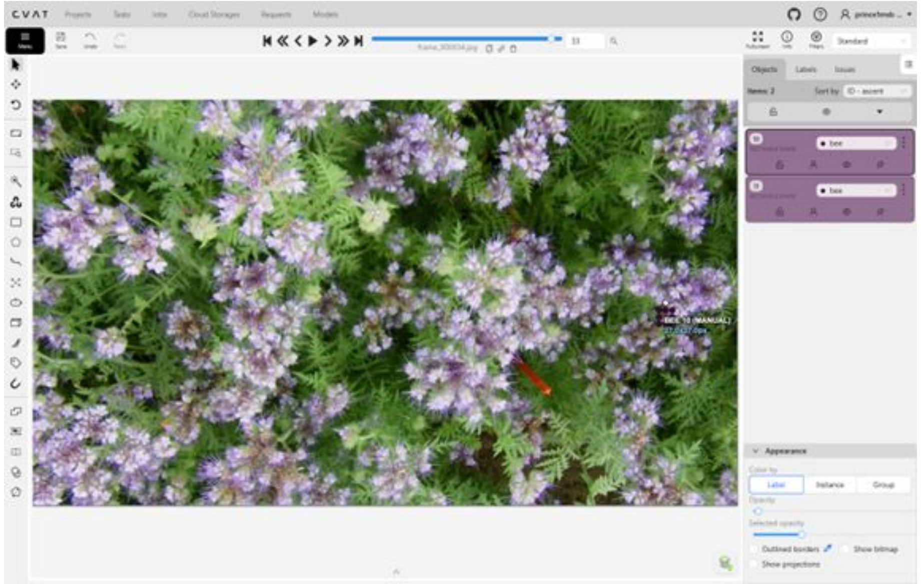
Major Problems



- misclassified flowers and background regions as insects
- Common failure cases included:
  - Flower centers
  - Bright vegetation
  - Background textures

Hard Negative Mining

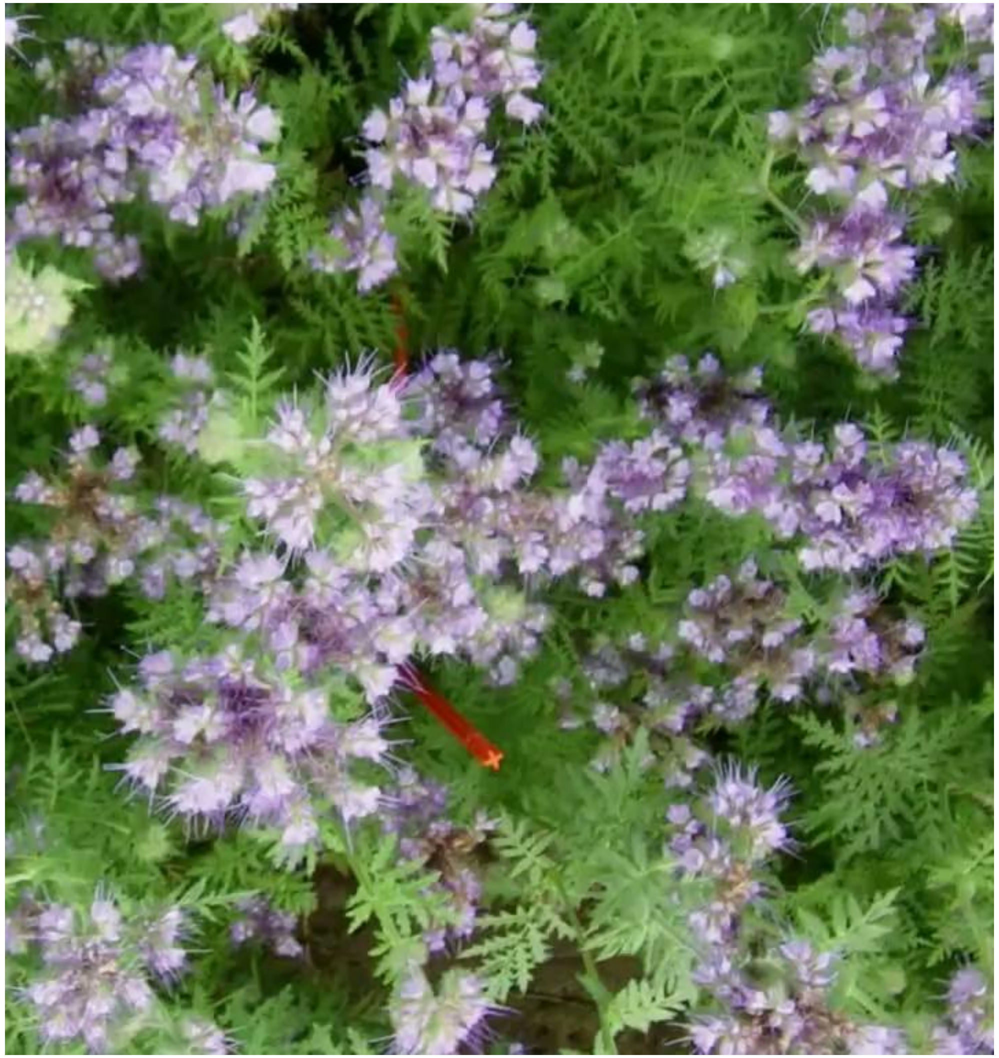
- Collected false positive detections.
- Manually annotated these regions as hard negatives
- Added the corrected samples to the training dataset.
- Retrained the detection models using the updated dataset.



YOLOv11L	New Value	old values
mAP@50-95	**0.389**	**0.275**
mAP@50	**0.428**	**0.421**
Precision	**0.732**	**0.638**
Recall	**0.393**	**0.413**

YOLO 11 (hardnegative)

- Reduced false positive detections.
- Improved model precision.
- Better discrimination between insects.
- Increased robustness in complex field environments.



Summary

- Initiated creating an additional dataset from raw 2024 field videos.
- Trained and evaluated multiple YOLO and RF-DETR models.
- Compared model performance using Precision, Recall and mAP.
- Identified and reduced false positives using hard negative mining.
- Future work includes tracking pipeline integration using ByteTrack/DeepSORT and monitoring statistics/dashboard.

Next Step

- 🚀 Fine-tune RF-DETR with hard negatives.
- 🔍 Use SAHI to improve detection.
- 🔄 Integrate ByteTrack/DeepSORT
- 🖼️ Improve the dataset with new images and videos.
- 📊 Develop a statistics dashboard.

Thank you

RF-DETR

Project Plan
(to be written but not to be presented)

Project Plan: Key Achievements Reached

- Goal: Describe the key achievements you have reached as a group. Also comment shortly, if plans have changed and why.

Tasks of Divyesh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)	Tasks of Person 5 (UniID)
- Trained the models and evaluate between each and everymodels.	- Literature review on bee detection and tracking methods - Compared YOLOv5, YOLOv7, and YOLOv8 models - Researched DeepSORT and ByteTrack tracking algorithms	- Studied environmental monitoring systems - Setup Python, OpenCV, and Jupyter environment - Worked on evaluation metrics and research documentation -started creating the dataset from raw video - Analyze tracking performance and movement patterns	- Studied insect and pollinator monitoring datasets - Researched and compared detection models for bee monitoring	List the performed tasks

Project Plan: Key Achievements Planned

- Goal: Describe the key achievements you have planned for the next period as a group.

Tasks of Diveysh (s54dkana)	Tasks of Prince (s57pbamb)	Tasks of Aniket (s17alimb)	Tasks of Rohit (s09rrath)	Tasks of Person 5 (UniID)
- Improve model accuracy using advanced methods - Implement class balancing techniques - Experiment with different detection algorithms - Optimize bee detection for small objects	- Apply data augmentation using Albumentations - Prepare train/validation/test datasets - Integrate DeepSORT and ByteTrack tracking - Improve video processing pipeline	- Research real-time deployment possibilities - Work on environmental analytics integration - Prepare final research documentation and reports	- Design and develop the Streamlit monitoring dashboard - Implement temporal bee activity charts - Write and compile the final project report	